How to model planetary fluids with the use of ML: Spherical Fourier Neural Operators

Spherical Fourier Neural Operators: Learning Stable Dynamics on the Sphere

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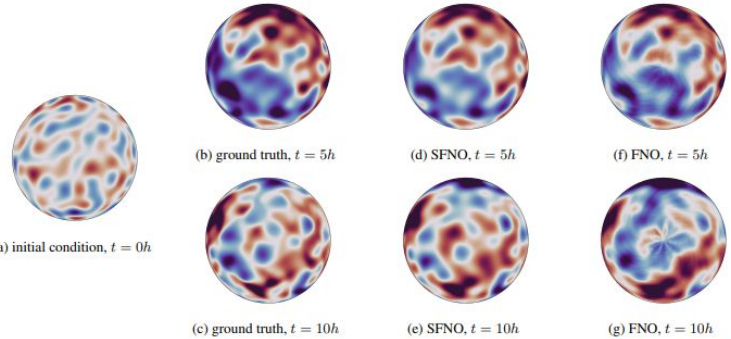

Figure 4. Solutions to the Shallow Water Equations on the rotating Sphere predicted by SFNO and FNO architectures in comparison to the ground truth solution computed using a classical spectral solver. Plots depict the geopotential height at 5 and 10 hours, corresponding to 5 and 10 autoregressive steps respectively. The view is centered on the south pole to highlight artifacts arising due to the non-geometrical treatment of the sphere.

- what is the problem
  a. What is SFNO and how does it work?
  b. How does mapping work from regular FNO to SFNO?
  c. How are temporal info stored at each grid point and across grid point?
  d. To what point would SFNO break and no longer hold for SWE?
- what approach do you plan to take
  a. Use torch-harmonics code to implement a toy SFNO to solve Shallow Water Equation (representative of a majority of atmospheric and oceanic fluids - sea surface temperatures)
- Evaluation
  a. Reproduce Results from Bonev et al. 2023
  b. Check outcome with changes to hyperparameters
- reflection and stretch goals
  a. Stretch Goals - compare with DeepONets